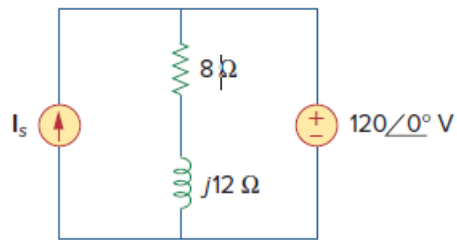


Problem

Determine I_s in the circuit of Fig. 11.83, if the voltage source supplies 6 kW and 1.2 kVAR (leading).

Figure 11.83:



Step-by-step solution

Step 1 of 4

Refer to Figure 11.83 in the text book.

Consider the circuit diagram shown in Figure 1.

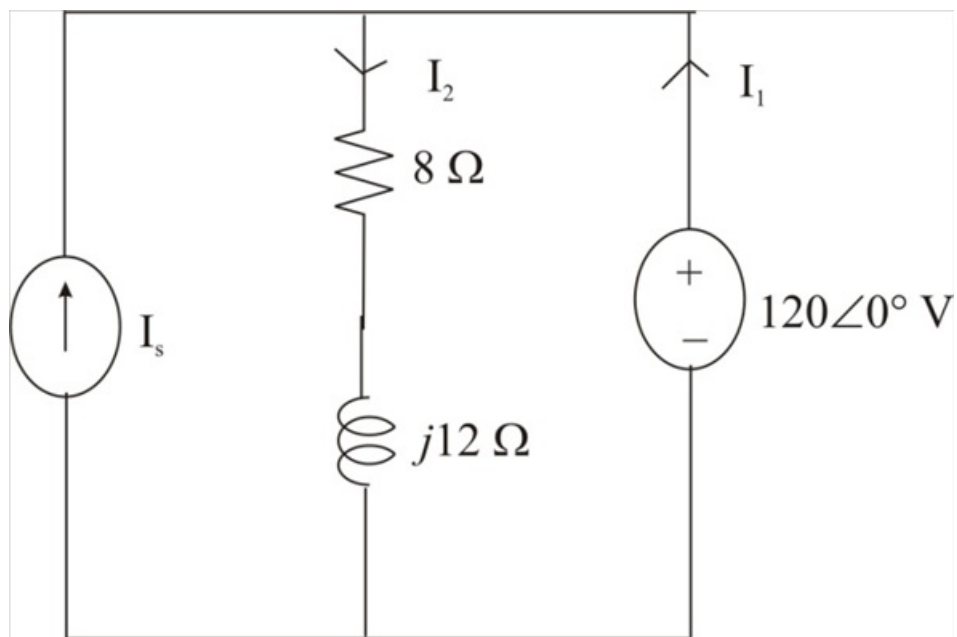


Figure 1

Step 2 of 4

The value of the complex power is S_1 is, $(6 \text{ k} - j1.2 \text{ k}) \text{ VA}$.

Calculate the value of the current I_1 .

$$I_1 = \left(\frac{S_1}{V} \right)^*$$

Substitute $120\angle 0^\circ \text{ V}$ for V and $(6 \text{ k} - j1.2 \text{ k}) \text{ VA}$ for S_1 .

$$\begin{aligned} I_1 &= \left(\frac{6 \text{ k} - j1.2 \text{ k}}{120\angle 0^\circ} \right)^* \\ &= \left(\frac{6.119\angle -11.31^\circ \text{ k}}{120\angle 0^\circ} \right)^* \\ &= (50.992\angle -11.31^\circ)^* \\ &= 50.992\angle 11.31^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_1 is $50.992\angle 11.31^\circ \text{ A}$.

Step 3 of 4

Calculate the value of the current I_2 .

$$\begin{aligned} I_2 &= \frac{120\angle 0^\circ}{8 + j12} \\ &= \frac{120\angle 0^\circ}{14.422\angle 56.31^\circ} \\ &= 8.327\angle -56.31^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is $8.327\angle -56.31^\circ \text{ A}$.

Step 4 of 4

Calculate the value of the current I_s .

$$I_s = I_2 - I_1$$

Substitute $8.327\angle -56.31^\circ \text{ A}$ for I_2 and $50.992\angle 11.31^\circ \text{ A}$ for I_1 .

$$\begin{aligned} I_s &= 8.327\angle -56.31^\circ - 50.992\angle 11.31^\circ \\ &= 48.437\angle -159.54^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_s is, $48.437\angle -159.54^\circ \text{ A}$.